

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

SlopeShield — Intelligent Landslide Risk & Safe Route System

Student / Team Information

Team Name: Team #	<i>The Debug Divas</i>
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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Hilly regions are highly vulnerable to landslides, especially during heavy rainfall. Landslides can block roads, damage infrastructure, and endanger travellers and emergency services. SlopeShield identifies landslide-prone areas using rainfall, slope, soil moisture, elevation, historical landslide records and road conditions. It also recommends safer routes by considering the risk of different road segments. This can improve travel safety, support emergency response and help users make better decisions during hazardous conditions.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

Landslide risk is currently assessed using hazard maps, government advisories and geological surveys. However these sources are often static and are not integrated with navigation systems. Most navigation apps focus on distance, traffic and travel time rather than landslide risk. SlopeShield addresses this gap by combining landslide risk assessment with safe route recommendation.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The main aim of SlopeShield is to check landslide risk and suggest safer routes using DSA and OOP concepts. The main steps are:

- 1. Collect and organize environmental and road data.*
- 2. Create a risk system and divide areas into risk levels.*
- 3. Make a weighted road graph using Dijkstra's algorithm and a min-heap priority queue.*
- 4. Create a web interface for showing risks and routes.*
- 5. Test the system and complete the final documentation.*

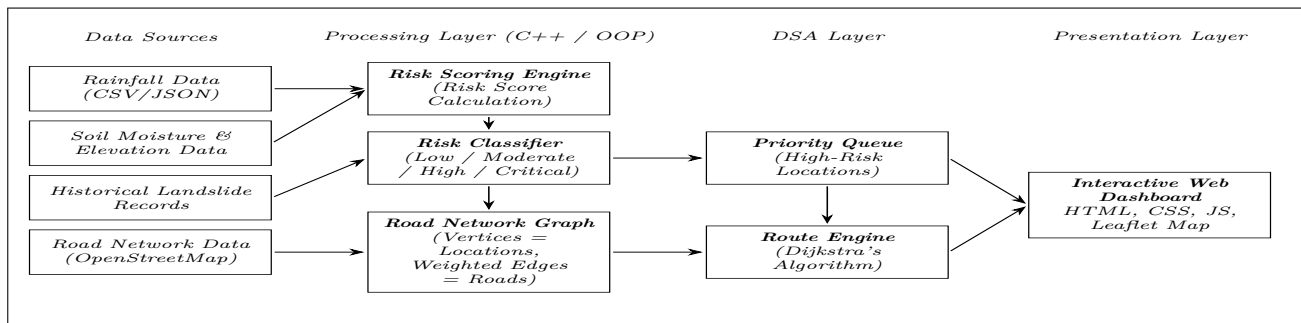
Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

SlopeShield will use C++ and OOP to model locations, roads, risk profiles and road network. Environmental data such as rainfall, slope, soil moisture, elevation, historical landslides and road conditions will be used to calculate risk levels. The road network will be represented as a weighted graph. Dijkstra's algorithm and a priority queue (min-heap) will be used to find safer routes while considering risk. A web interface using HTML, CSS, JavaScript, Leaflet and OpenStreetMap will display risk zones and recommended routes. Git and GitHub support version control and collaboration.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The main deliverable is a web-based SlopeShield system that identifies landslide prone areas, displays risk levels and recommends safer routes between locations.

The project will include a C++ backend with OOP and DSA implementations, a road network graph, risk scoring module, Dijkstra's algorithm, priority queue, web dashboard, sample datasets, documentation and a final demonstration.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

We assume that reliable historical and geographical data such as rainfall, elevation, soil moisture and past landslide records are available in CSV/JSON format. Road network data is assumed to be available from OpenStreetMap. The initial system will use periodically updated datasets rather than real-time sensor data.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. OpenStreetMap contributors – OpenStreetMap.
2. Leaflet.js documentation – Leaflet.
3. Cormen, Leiserson, Rivest, and Stein – Introduction to Algorithms, for graph, priority queue and Dijkstra's algorithm.
4. Relevant landslide hazard and geological survey publications used for background research on landslide risk factors.